Body Fluids and Circulation

15.0 Why Body Fluids Exist - The Chapter's Starting Point

All living cells have to be provided with **nutrients, O_2 and other essential substances**. Also, the **waste or harmful substances** produced have to be **removed continuously** for healthy functioning of tissues. It is therefore essential to have **efficient mechanisms** for the movement of these substances **to the cells and from the cells**.

Different groups of animals have evolved **different methods** for this transport.

- **Simple organisms** like **sponges and coelenterates** circulate **water from their surroundings** through their **body cavities** to facilitate the cells to exchange these substances.
- **More complex organisms** use **special fluids within their bodies** to transport such materials.

Blood is the **most commonly used body fluid** by most of the higher organisms, including humans, for this purpose. Another body fluid, **lymph**, also helps in the transport of certain substances.

> ⓘ *[NOTE] This chapter covers the **composition and properties of blood and lymph (tissue fluid)**, and the **mechanism of circulation of blood**.*

Chapter contents (p. 193 margin panel):

15.1 Blood

- **Blood** is a special **connective tissue** consisting of a **fluid matrix, plasma**, and **formed elements**.

> ⓘ ***[NOTE] Why blood counts as a connective tissue** (assumed by Exercise 4; the NCERT body only asserts it): a connective tissue is **mesodermal in origin** and consists of **cells scattered in an extracellular matrix**. Blood fits exactly **-** its **formed elements** are the cells, and **plasma** is the matrix, which in blood happens to be **fluid** rather than solid or jelly-like.*

15.1.1 Plasma

Plasma is a **straw coloured, viscous fluid** constituting nearly **55 per cent** of the blood.

- **90-92 per cent** of plasma is **water**; **proteins** contribute **6-8 per cent** of it.
- **Fibrinogen, globulins and albumins** are the **major proteins**:
  - **Fibrinogens -** needed for **clotting or coagulation** of blood.
  - **Globulins -** primarily involved in **defense mechanisms** of the body.
  - **Albumins -** help in **osmotic balance**.
- Plasma also contains **small amounts of minerals** like Na^+, Ca^{++}, Mg^{++}, HCO_3^-, Cl^-, etc.
- **Glucose, amino acids, lipids**, etc., are also present in the plasma as they are **always in transit** in the body.
- **Factors for coagulation or clotting** of blood are also present in the plasma in an **inactive form**.

- **Serum:** plasma **without the clotting factors**.

15.1.2 Formed Elements

Erythrocytes, leucocytes and platelets are collectively called **formed elements** (Figure 15.1) and they constitute nearly **45 per cent** of the blood.

15.1.2a Erythrocytes (RBC)

- **Erythrocytes** or **red blood cells (RBC)** are the **most abundant** of all the cells in blood.
- A healthy adult man has, on an average, **5 millions to 5.5 millions of RBCs mm^{-3}** of blood.
- RBCs are formed in the **red bone marrow** in the adults.

- RBCs are **devoid of nucleus** in **most of the mammals** and are **biconcave** in shape.
- They have a **red coloured, iron containing complex protein** called **haemoglobin**, hence the colour and name of these cells.
- A healthy individual has **12-16 gms of haemoglobin** in every **100 mL** of blood. These molecules play a significant role in **transport of respiratory gases**.
- RBCs have an average **life span of 120 days**, after which they are destroyed in the **spleen** (**graveyard of RBCs**).

15.1.2b Leucocytes (WBC)

- **Leucocytes** are also known as **white blood cells (WBC)** as they are **colourless** due to the **lack of haemoglobin**.
- They are **nucleated** and are **relatively lesser in number**, which averages **6000-8000 mm^{-3}** of blood.
- Leucocytes are **generally short lived**.
- Two main categories of WBCs - **granulocytes** and **agranulocytes**. **Neutrophils, eosinophils and basophils** are the different types of **granulocytes**, while **lymphocytes and monocytes** are the **agranulocytes**.

WBC type	Group	Per cent of total WBCs	Function
Neutrophils	Granulocyte	60-65 per cent (most abundant)	Phagocytic - destroy foreign organisms entering the body
Eosinophils	Granulocyte	2-3 per cent	Resist infections; also associated with allergic reactions
Basophils	Granulocyte	0.5-1 per cent (least)	Secrete histamine, serotonin, heparin, etc.; involved in inflammatory reactions
Lymphocytes ('B' and 'T' forms)	Agranulocyte	20-25 per cent	Both B and T lymphocytes are responsible for immune responses of the body
Monocytes	Agranulocyte	6-8 per cent	Phagocytic - destroy foreign organisms entering the body

☆ ***[MEMORY AID - not in NCERT] Never Let Mature Erythrocytes Be** = Neutrophils 60-65, Lymphocytes 20-25, Monocytes 6-8, Eosinophils 2-3, Basophils 0.5-1 - the five WBCs in **descending order of abundance**.*

15.1.2c Platelets (Thrombocytes)

- **Platelets**, also called **thrombocytes**, are **cell fragments** produced from **megakaryocytes** (special cells in the bone marrow).
- Blood normally contains **1,500,00-3,500,00 platelets mm^{-3}**.
- Platelets can release a **variety of substances**, most of which are involved in the **coagulation or clotting** of blood.
- A **reduction in their number** can lead to **clotting disorders**, which will lead to **excessive loss of blood** from the body.

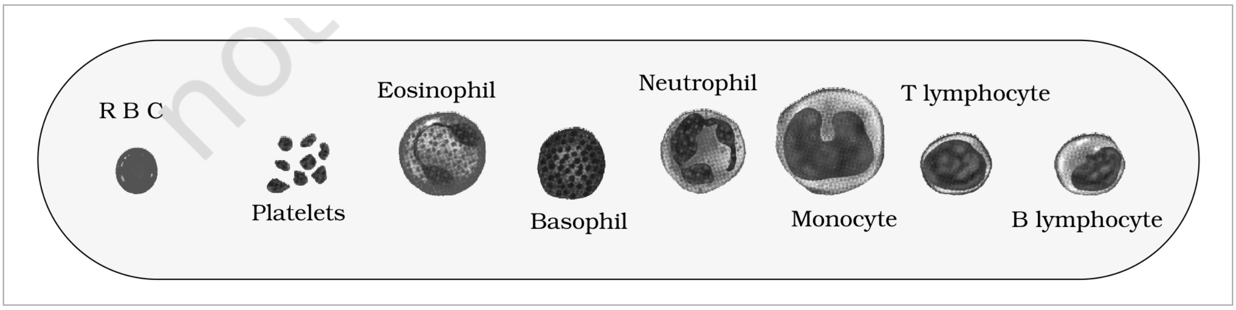

Figure 15.1 Diagrammatic representation of formed elements in blood

ⓘ ***[NOTE] Figure 15.1 callouts, verbatim:** R B C; Platelets; Eosinophil; Basophil; Neutrophil; Monocyte; T lymphocyte; B lymphocyte.*

15.1.3 Blood Groups

Blood of human beings **differ in certain aspects** though it appears to be similar. **Various types of grouping** of blood has been done. Two such groupings - the **ABO** and **Rh** - are **widely used all over the world**.

15.1.3.1 ABO grouping

ABO grouping is based on the **presence or absence of two surface antigens** (chemicals that can induce immune response) on the RBCs, namely **A** and **B**.

- Similarly, the **plasma** of different individuals contain **two natural antibodies** (proteins produced in response to antigens).

The distribution of **antigens and antibodies** in the four groups of blood **- A, B, AB and O -** and the **donor's compatibility** are given in Table 15.1.

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

TABLE 15.1 Blood Groups and Donor Compatibility

> *[NOTE] During **blood transfusion**, **any blood cannot be used**; the blood of a **donor** has to be **carefully matched** with the blood of a **recipient** before any blood transfusion, to avoid severe problems of **clumping (destruction of RBC)**.*

- Group **'O'** blood can be donated to persons with **any other blood group**, hence 'O' group individuals are called **universal donors**.
- Persons with **'AB'** group can accept blood from persons with **AB as well as the other groups** of blood. Therefore, such persons are called **universal recipients**.

15.1.3.2 Rh grouping

Another antigen, the **Rh antigen** similar to one present in **Rhesus monkeys** (hence **Rh**), is also observed on the surface of RBCs of **majority (nearly 80 per cent)** of humans. This grouping is therefore done on the presence or absence of another antigen called the **Rhesus factor (Rh)** on the surface of RBCs.

- Such individuals are called **Rh positive (Rh+ve)** and those in whom this antigen is **absent** are called **Rh negative (Rh-ve)**.
- An **Rh-ve person**, if **exposed to Rh+ve blood**, will form **specific antibodies** against the Rh antigens. Therefore, **Rh group should also be matched** before transfusions.

15.1.3.2a Erythroblastosis foetalis

A **special case of Rh incompatibility (mismatching)** has been observed between the **Rh-ve blood of a pregnant mother** with **Rh+ve blood of the foetus**.

1. Rh antigens of the foetus **do not get exposed** to the Rh-ve blood of the mother in the **first pregnancy**, as the two bloods are **well separated by the placenta**.
2. However, **during the delivery of the first child**, there is a possibility of **exposure of the maternal blood** to **small amounts of the Rh+ve blood** from the foetus.
3. In such cases, the mother **starts preparing antibodies** against Rh antigen in her blood.
4. In case of her **subsequent pregnancies**, the **Rh antibodies from the mother (Rh-ve)** can **leak into the blood of the foetus (Rh+ve)** and **destroy the foetal RBCs**.
5. This could be **fatal to the foetus**, or could cause **severe anaemia and jaundice** to the baby.

- This condition is called **erythroblastosis foetalis**.

> *[NOTE] This can be **avoided** by administering **anti-Rh antibodies** to the mother **immediately after the delivery of the first child**.*

15.1.4 Coagulation of Blood

When you **cut your finger or hurt yourself**, your wound **does not continue to bleed for a long time**; usually the blood **stops flowing after sometime**. Blood exhibits **coagulation or clotting** in response to an **injury or trauma**.

This is a mechanism to **prevent excessive loss of blood** from the body.

You would have observed a **dark reddish brown scum** formed at the site of a cut or an injury over a period of time.

- It is a **clot** or **coagulam** formed mainly of a **network of threads called fibrins**, in which **dead and damaged formed elements** of blood are trapped.

1. An **injury or a trauma stimulates the platelets** in the blood to release certain **factors** which **activate the mechanism of coagulation**. Certain factors released by the **tissues at the site of injury** also can initiate coagulation.
2. A **series of linked enzymic reactions (cascade process)**, involving a number of factors present in the plasma in an **inactive state**, forms an **enzyme complex, thrombokinase**.
3. **Thrombokinase** is required to form **thrombin** from another **inactive substance** present in the plasma called **prothrombin**.
4. **Fibrins** are formed by the conversion of **inactive fibrinogens** in the plasma by the enzyme **thrombin**.

*[NOTE] **Calcium ions** play a **very important role** in clotting.*

*[MEMORY AID - not in NCERT] Read the cascade **backwards** to remember it: **fibrin** needs **thrombin**, thrombin needs **thrombokinase**, thrombokinase needs the **cascade + platelet factors**. Each step turns an **inactive plasma factor** into its active form.*

15.2 Lymph (Tissue Fluid)

As the blood passes through the **capillaries in tissues**, **some water along with many small water soluble substances move out** into the **spaces between the cells** of tissues, **leaving the larger proteins and most of the formed elements** in the blood vessels.

- This fluid released out is called the **interstitial fluid** or **tissue fluid**. It has the **same mineral distribution** as that in plasma.
  - **Exchange of nutrients, gases**, etc., between the **blood and the cells** always occur **through this fluid**.
  - An **elaborate network of vessels** called the **lymphatic system** collects this fluid and **drains it back to the major veins**.
- The fluid present in the lymphatic system is called the **lymph**. Lymph is a **colourless fluid** containing **specialised lymphocytes**, which are responsible for the **immune responses** of the body.
  - Lymph is also an **important carrier for nutrients, hormones**, etc.
  - **Fats are absorbed through lymph** in the **lacteals** present in the **intestinal villi**.

*[NOTE] Lymph is **almost similar to blood except for the protein content and the formed elements**.*

Feature	Blood	Lymph
Colour	Red (haemoglobin in RBCs)	Colourless
Proteins	High - fibrinogen, globulins, albumins all present	Low - larger proteins are left behind in the blood vessels
Formed elements	RBCs, all types of WBCs and platelets present	No RBCs and no platelets; contains specialised lymphocytes only
Flows in	Closed network of arteries, capillaries and veins	Lymphatic system, draining back into the major veins
Main function	Transport of respiratory gases, nutrients, wastes, hormones	Immune responses; carrier of nutrients and hormones; absorption of fats in the lacteals

Blood versus lymph - the contrast assumed by Exercises 5 and 7(a).

15.3 Circulatory Pathways

The **circulatory patterns** are of **two types - open or closed**.

- **Open circulatory system** is present in **arthropods and molluscs**, in which blood pumped by the heart passes through **large vessels into open spaces or body cavities called sinuses**.
- **Annelids and chordates** have a **closed circulatory system**, in which the blood pumped by the heart is **always circulated through a closed network of blood vessels**. This pattern is considered to be **more advantageous**, as the **flow of fluid can be more precisely regulated**.

> ⓘ ***[NOTE] All vertebrates and a few invertebrates*** *have a* ***closed circulatory system****.*

15.3a Evolution of the Vertebrate Heart

All vertebrates possess a muscular chambered heart.

Group	Heart chambers	Circulation
Fishes	**2-chambered** - an atrium and a ventricle	**Single circulation.** The heart pumps out deoxygenated blood, which is oxygenated by the **gills** and supplied to the body parts, from where deoxygenated blood is returned to the heart.
Amphibians and reptiles (except crocodiles)	**3-chambered** - two atria and a single ventricle	**Incomplete double circulation.** The left atrium receives oxygenated blood from the gills / lungs / skin and the right atrium gets the deoxygenated blood from other body parts; however, they **get mixed up in the single ventricle**, which pumps out **mixed blood**.
Crocodiles, birds and mammals	**4-chambered** - two atria and two ventricles	**Double circulation.** Oxygenated and deoxygenated blood received by the left and right atria respectively passes on to the ventricles of the same sides. The ventricles pump it out **without any mixing up**, i.e., **two separate circulatory pathways** are present.

> ☆ ***[MEMORY AID - not in NCERT] 2 - 3 - 4****: fishes 2, amphibians and reptiles 3, birds and mammals 4. The two exceptions both sit with the 4s:* ***crocodiles*** *are reptiles with a 4-chambered heart.*

15.3.1 Human Circulatory System

- **Human circulatory system**, also called the **blood vascular system**, consists of a **muscular chambered heart**, a **network of closed branching blood vessels** and **blood**, the fluid which is circulated.
  - **Heart**, the **mesodermally derived** organ, is situated in the **thoracic cavity**, **in between the two lungs**, **slightly tilted to the left**. It has the **size of a clenched fist**.
  - It is protected by a **double walled membranous bag, pericardium**, enclosing the **pericardial fluid**.

15.3.1a Chambers, Septa and Valves

- Our heart has **four chambers -** two relatively **small upper chambers called atria** and two **larger lower chambers called ventricles**.
- A **thin, muscular wall** called the **inter-atrial septum** separates the right and the left atria, whereas a **thick-walled inter-ventricular septum** separates the left and the right ventricles (Figure 15.2).
- The atrium and the ventricle of the same side are also separated by a **thick fibrous tissue called the atrio-ventricular septum**. However, **each of these septa are provided with an opening** through which the two chambers of the same side are connected.
- The opening between the **right atrium and the right ventricle** is guarded by a valve formed of **three muscular flaps or cusps, the tricuspid valve**, whereas a **bicuspid or mitral valve** guards the opening between the **left atrium and the left ventricle**.
- The openings of the **right and the left ventricles** into the **pulmonary artery** and the **aorta** respectively are provided with the **semilunar valves**.

> ⓘ ***[NOTE]*** *The valves in the heart allow the flow of blood* ***only in one direction****, i.e.,* ***from the atria to the ventricles*** *and* ***from the ventricles to the pulmonary artery or aorta****. These valves* ***prevent any backward flow****.*

- The **entire heart is made of cardiac muscles**. The **walls of ventricles are much thicker** than that of the atria.

15.3.1b The Nodal Tissue

A **specialised cardiac musculature** called the **nodal tissue** is also distributed in the heart (Figure 15.2).

- A patch of this tissue is present in the **right upper corner of the right atrium** called the **sino-atrial node (SAN)**.

- Another mass of this tissue is seen in the **lower left corner of the right atrium**, close to the atrio-ventricular septum, called the **atrio-ventricular node (AVN)**.
- A bundle of nodal fibres, the **atrio-ventricular bundle (AV bundle)**, continues from the AVN, which passes through the atrio-ventricular septa to emerge on the top of the inter-ventricular septum and **immediately divides into a right and left bundle**.
- These branches give rise to **minute fibres throughout the ventricular musculature** of the respective sides and are called **purkinje fibres**.

● The nodal musculature has the ability to **generate action potentials without any external stimuli**, i.e., it is **autoexcitable**.

- However, the **number of action potentials** that could be generated in a minute **vary at different parts** of the nodal system. The **SAN can generate the maximum number of action potentials, i.e., 70-75 min^{-1}**, and is responsible for **initiating and maintaining the rhythmic contractile activity** of the heart. Therefore, it is called the **pacemaker**.
- Our heart normally beats **70-75 times in a minute** (average **72 beats min^{-1}**).

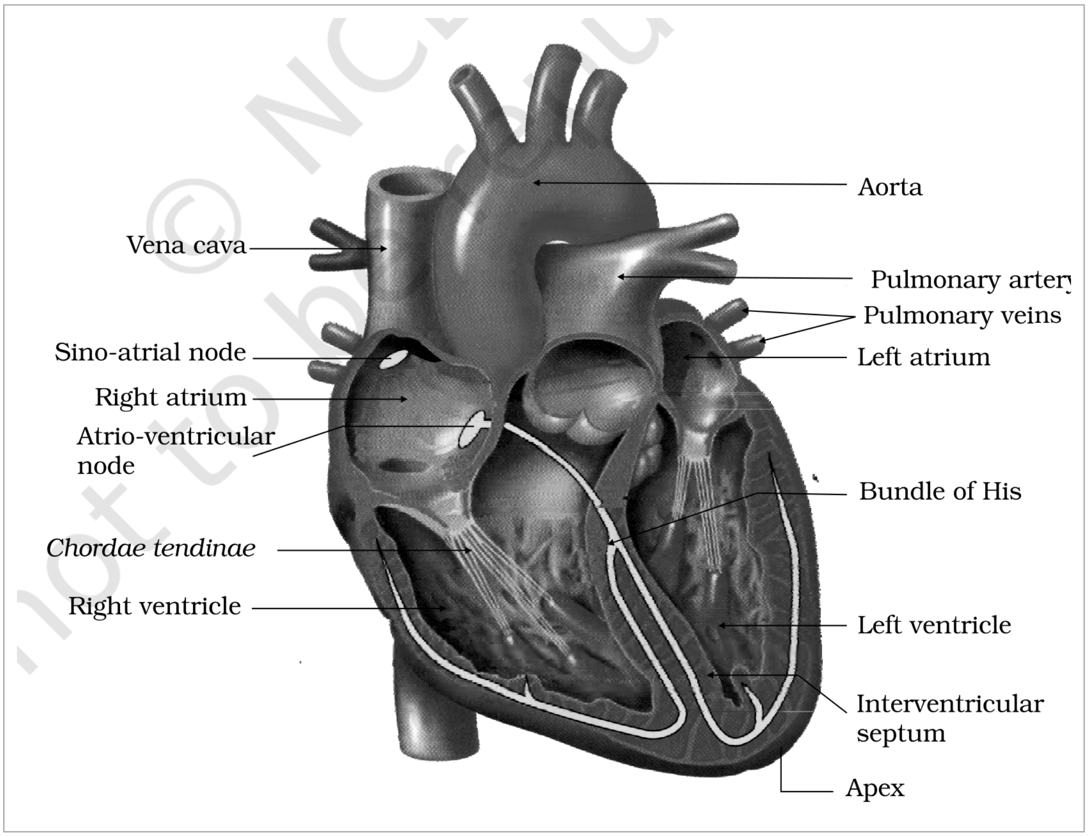

Figure 15.2 Section of a human heart

> ⓘ ***[NOTE] Figure 15.2 callouts, verbatim:*** *Vena cava; Sino-atrial node; Right atrium; Atrio-ventricular node; Chordae tendinae; Right ventricle; Aorta; Pulmonary artery; Pulmonary veins; Left atrium; Bundle of His; Left ventricle; Interventricular septum; Apex.*

15.3.2 Cardiac Cycle

How does the heart function? Let us take a look.

▲ *(cycle - last step feeds back to step 1)*

1 To begin with, **all the four chambers** of heart are in a **relaxed state**, i.e., they are in **joint diastole**. As the **tricuspid and bicuspid valves are open**, blood from the **pulmonary veins and vena cava** flows into the **left and the right ventricle** respectively through the left and right atria. The **semilunar valves are closed** at this stage.

2 The **SAN now generates an action potential** which stimulates **both the atria** to undergo a **simultaneous contraction - the atrial systole**. This **increases the flow of blood into the ventricles by about 30 per cent**.

3 The action potential is conducted to the ventricular side by the **AVN and AV bundle**, from where the **bundle of His** transmits it through the **entire ventricular musculature**.

4 This causes the ventricular muscles to contract (**ventricular systole**); the atria undergoes **relaxation (diastole)**, coinciding with the ventricular systole.

5 Ventricular systole **increases the ventricular pressure**, causing the **closure of tricuspid and bicuspid valves** due to **attempted backflow** of blood into the atria.

6 As the ventricular pressure **increases further**, the **semilunar valves** guarding the **pulmonary artery (right side)** and the **aorta (left side)** are **forced open**, allowing the blood in the ventricles to flow through these vessels into the circulatory pathways.

7 The ventricles now **relax (ventricular diastole)** and the **ventricular pressure falls**, causing the **closure of semilunar valves**, which **prevents the backflow** of blood into the ventricles.

8 As the ventricular pressure **declines further**, the **tricuspid and bicuspid valves are pushed open** by the pressure in the atria exerted by the blood which was being **emptied into them by the veins**.

9 The blood now once again **moves freely to the ventricles**. The ventricles and atria are now again in a **relaxed (joint diastole)** state, as earlier. Soon the **SAN generates a new action potential** and the events described above are **repeated in that sequence** and the process continues.

- This **sequential event in the heart which is cyclically repeated** is called the **cardiac cycle**, and it consists of **systole and diastole of both the atria and ventricles**.

15.3.2a Stroke Volume and Cardiac Output

- As mentioned earlier, the heart beats **72 times per minute**, i.e., that many cardiac cycles are performed per minute. From this it could be deduced that the **duration of a cardiac cycle is 0.8 seconds**.
- During a cardiac cycle, each ventricle pumps out approximately **70 mL of blood**, which is called the **stroke volume** (also called the **stroke or beat volume**).
- The **stroke volume multiplied by the heart rate** (no. of beats per min.) gives the **cardiac output**. The cardiac output can be defined as the **volume of blood pumped out by each ventricle per minute**, and averages **5000 mL or 5 litres** in a healthy individual.

> ⓘ ***[NOTE]*** *The body has the ability to **alter the stroke volume as well as the heart rate**, and thereby the **cardiac output**. For example, the cardiac output of an **athlete** will be **much higher** than that of an **ordinary man**.*

15.3.2b Heart Sounds

During each cardiac cycle **two prominent sounds** are produced, which can be easily heard through a **stethoscope**.

- The **first heart sound (lub)** is associated with the **closure of the tricuspid and bicuspid valves**.
- The **second heart sound (dub)** is associated with the **closure of the semilunar valves**.

> ⓘ ***[NOTE]*** *These sounds are of **clinical diagnostic significance**.*

15.3.3 Electrocardiogram (ECG)

You are probably familiar with this scene from a typical hospital television show: a **patient is hooked up to a monitoring machine** that shows **voltage traces** on a screen and makes the sound '... pip... pip... pip..... peeeeeeeeeeeeeeeeeeeeee' as the patient goes into **cardiac arrest**. This type of machine (**electro-cardiograph**) is used to obtain an **electrocardiogram (ECG)**.

- **ECG** is a **graphical representation of the electrical activity of the heart** during a **cardiac cycle**.
- To obtain a **standard ECG** (as shown in Figure 15.3), a patient is connected to the machine with **three electrical leads** (**one to each wrist and to the left ankle**) that **continuously monitor the heart activity**.
- For a **detailed evaluation** of the heart's function, **multiple leads** are attached to the **chest region**. Here, we will talk **only about a standard ECG**.

- Each **peak** in the ECG is identified with a **letter from P to T** that corresponds to a **specific electrical activity** of the heart.

1. The **P-wave** represents the **electrical excitation (or depolarisation) of the atria**, which leads to the **contraction of both the atria**.
2. The **QRS complex** represents the **depolarisation of the ventricles**, which **initiates the ventricular contraction**. The contraction starts **shortly after Q** and marks the **beginning of the systole**.
3. The **T-wave** represents the **return of the ventricles from excited to normal state (repolarisation)**. The **end of the T-wave** marks the **end of systole**.

> ⓘ ***[NOTE]*** *By* ***counting the number of QRS complexes*** *that occur in a given time period, one can* ***determine the heart beat rate*** *of an individual.*

> ⓘ ***[NOTE]*** *Since the ECGs obtained from different individuals have* ***roughly the same shape*** *for a given lead configuration,* ***any deviation from this shape indicates a possible abnormality or disease****. Hence, it is of a* ***great clinical significance****.*

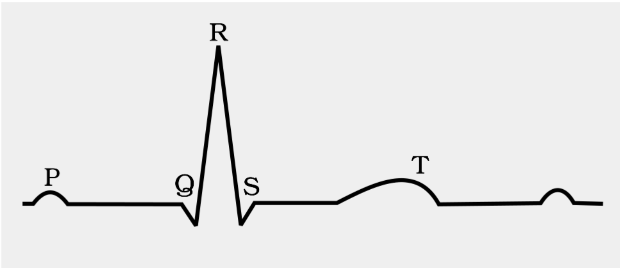

Figure 15.3 Diagrammatic presentation of a standard ECG

> ⓘ ***[NOTE] Figure 15.3 callouts, verbatim:*** *P; Q; R; S; T.*

15.4 Double Circulation

The blood flows **strictly by a fixed route** through **Blood Vessels - the arteries and veins**.

- Basically, **each artery and vein consists of three layers**: an **inner lining of squamous endothelium, the tunica intima**; a **middle layer of smooth muscle and elastic fibres, the tunica media**; and an **external layer of fibrous connective tissue with collagen fibres, the tunica externa**.
- The **tunica media is comparatively thin in the veins** (Figure 15.4).

As mentioned earlier, the blood pumped by the **right ventricle** enters the **pulmonary artery**, whereas the **left ventricle** pumps blood into the **aorta**.

1. **Pulmonary circulation.** The **deoxygenated blood** pumped into the **pulmonary artery** is passed on to the **lungs**, from where the **oxygenated blood** is carried by the **pulmonary veins** into the **left atrium**.
2. **Systemic circulation.** The **oxygenated blood** entering the **aorta** is carried by a network of **arteries, arterioles and capillaries** to the tissues, from where the **deoxygenated blood** is collected by a system of **venules, veins and vena cava** and emptied into the **right atrium** (Figure 15.4).

- The **systemic circulation** provides **nutrients, O_2 and other essential substances** to the tissues and takes **CO_2 and other harmful substances** away for **elimination**.

> ⓘ ***[NOTE]*** *We have a* ***complete double circulation****, i.e.,* ***two circulatory pathways****, namely,* ***pulmonary and systemic****, are present.* ***Its significance*** *(assumed by Exercise 6): because the two pathways are* ***fully separate****, oxygenated and deoxygenated blood* ***never mix****, so the tissues always receive* ***fully oxygenated blood -*** *which supports the high metabolic rate of birds and mammals.*

15.4a Two Special Vascular Connections

- A **unique vascular connection** exists between the **digestive tract and liver** called the **hepatic portal system**. The **hepatic portal vein** carries blood **from intestine to the liver** before it is delivered to the **systemic**

circulation.

- A special **coronary system** of blood vessels is present in our body **exclusively for the circulation of blood to and from the cardiac musculature**.

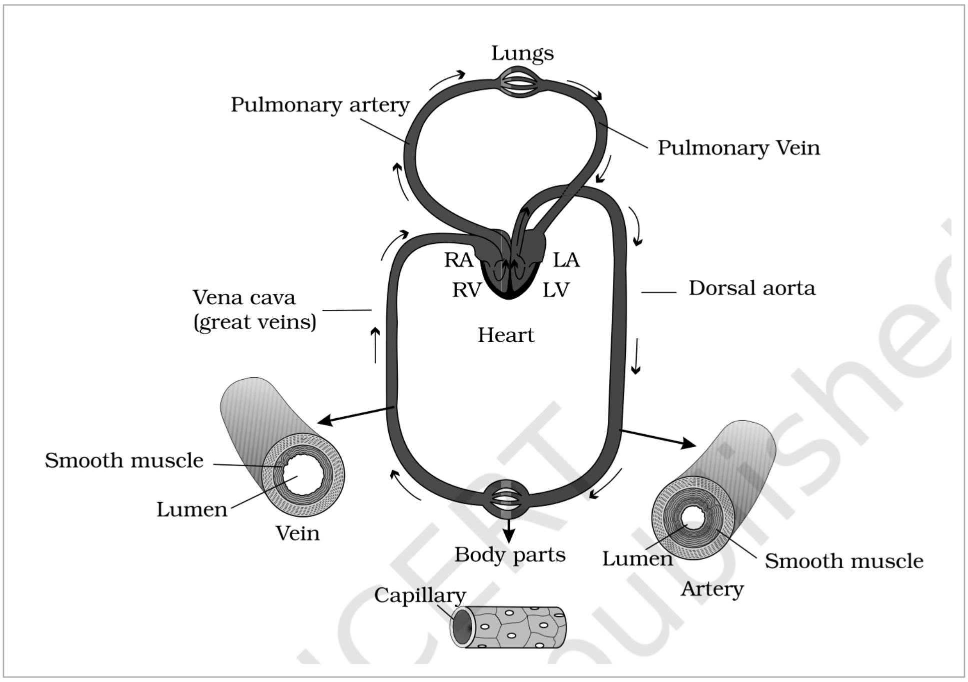

Figure 15.4 Schematic plan of blood circulation in human

> ⓘ ***[NOTE] Figure 15.4 callouts, verbatim:*** *Lungs; Pulmonary artery; Pulmonary Vein; Vena cava (great veins); RA; RV; LA; LV; Heart; Dorsal aorta; Body parts; Smooth muscle; Lumen; Vein; Capillary; Artery.*

15.5 Regulation of Cardiac Activity

- Normal activities of the heart are **regulated intrinsically**, i.e., **auto regulated by specialised muscles (nodal tissue)**, hence the heart is called **myogenic**.
  - A **special neural centre in the medulla oblangata** can **moderate the cardiac function** through the **autonomic nervous system (ANS)**.
  - Neural signals through the **sympathetic nerves** (part of ANS) can **increase** the **rate of heart beat**, the **strength of ventricular contraction** and thereby the **cardiac output**.
  - On the other hand, **parasympathetic neural signals** (another component of ANS) **decrease** the **rate of heart beat**, the **speed of conduction of action potential** and thereby the **cardiac output**.
  - **Adrenal medullary hormones** can also **increase the cardiac output**.

15.6 Disorders of Circulatory System

15.6a High Blood Pressure (Hypertension)

- **Hypertension** is the term for **blood pressure that is higher than normal (120/80)**.
  - In this measurement **120 mm Hg** (millimetres of mercury pressure) is the **systolic, or pumping, pressure** and **80 mm Hg** is the **diastolic, or resting, pressure**.
  - If repeated checks of blood pressure of an individual is **140/90 (140 over 90) or higher**, it shows **hypertension**.

> ⓘ ***[NOTE]*** *High blood pressure* ***leads to heart diseases*** *and also* ***affects vital organs like brain and kidney****.*

15.6b Coronary Artery Disease (CAD)

- **Coronary Artery Disease**, often referred to as **atherosclerosis**, affects the **vessels that supply blood to the heart muscle**.
  - It is caused by **deposits of calcium, fat, cholesterol and fibrous tissues**, which makes the **lumen of arteries narrower**.

15.6c Angina

- It is also called **'angina pectoris'**. A symptom of **acute chest pain** appears when **no enough oxygen is reaching the heart muscle**.
  - Angina can occur in **men and women of any age**, but it is **more common among the middle-aged and elderly**.
  - It occurs due to **conditions that affect the blood flow**.

15.6d Heart Failure

- **Heart failure** means the **state of heart when it is not pumping blood effectively enough to meet the needs of the body**.
  - It is sometimes called **congestive heart failure**, because **congestion of the lungs** is one of the **main symptoms** of this disease.

> ⓘ ***[NOTE]** Heart failure is **not the same as**:*
> ***cardiac arrest -** when the **heart stops beating**; or*
> ***a heart attack -** when the **heart muscle is suddenly damaged by an inadequate blood supply**.*

Recap Summary

Vertebrates circulate blood, a fluid connective tissue, in their body, to **transport essential substances to the cells** and to **carry waste substances from there**.

- Another fluid, **lymph (tissue fluid)**, is also used for the transport of certain substances.
- Blood comprises of a **fluid matrix, plasma** and **formed elements - RBCs (erythrocytes), WBCs (leucocytes) and platelets (thrombocytes)**.
- Human blood is grouped into the **ABO** system (based on the surface antigens **A** and **B** on RBCs) and the **Rh** system (based on the **Rhesus factor**).
- Our circulatory system consists of a **muscular pumping organ, the heart** (two atria and two ventricles), a **network of vessels** and the fluid, **blood**.
- **Cardiac musculature is auto-excitable.** The **SAN** generates the maximum number of action potentials per minute (**70-75 min^{-1}**), hence it is called the **pacemaker**.
- The action potential causes the **atria** and then the **ventricles** to undergo **contraction (systole)** followed by their **relaxation (diastole)**. The systole forces the blood to move **from the atria to the ventricles** and **to the pulmonary artery and the aorta**.
- The **cardiac cycle** is formed by these sequential events, cyclically repeated; a healthy person shows **72 such cycles per minute**. About **70 mL** is the **stroke (beat) volume**; the **cardiac output** averages **5 litres** per minute.
- The electrical activity of the heart can be recorded from the body surface using an **electrocardiograph**; the recording is the **electrocardiogram (ECG)**, which is of **clinical importance**.
- We have a **complete double circulation**: **pulmonary circulation** starts with the right ventricle pumping deoxygenated blood to the lungs, and **systemic circulation** starts with the left ventricle pumping oxygenated blood into the aorta.
- Though the heart is **autoexcitable**, its functions can be **moderated by neural and hormonal mechanisms**.

Ex Exercises - and Where Each Is Answered

#	Exercise question	Answered in
1	Name the components of the formed elements in the blood and mention one major function of each of them.	15.1.2 (RBC, WBC table, platelets) + Figure 15.1
2	What is the importance of plasma proteins?	15.1.1 - fibrinogen (clotting), globulins (defense), albumins (osmotic balance)
3	Match Column I with Column II. Column I: (a) Eosinophils (b) RBC (c) AB Group (d) Platelets (e) Systole. Column II: (i) Coagulation (ii) Universal Recipient (iii) Resist Infections (iv) Contraction of Heart (v) Gas transport	(a)-(iii) 15.1.2b; (b)-(v) 15.1.2a; (c)-(ii) 15.1.3.1; (d)-(i) 15.1.2c; (e)-(iv) 15.3.2
4	Why do we consider blood as a connective tissue?	15.1 - fluid matrix (plasma) plus formed elements, mesodermal in origin

#	Exercise question	Answered in
5	What is the difference between lymph and blood?	15.2 - blood versus lymph table
6	What is meant by double circulation? What is its significance?	15.4 - pulmonary and systemic pathways; significance stated in the NOTE
7	Write the differences between: (a) Blood and Lymph (b) Open and Closed system of circulation (c) Systole and Diastole (d) P-wave and T-wave	(a) 15.2 table; (b) 15.3; (c) 15.3.2; (d) 15.3.3
8	Describe the evolutionary change in the pattern of heart among the vertebrates.	15.3a - the 2 / 3 / 4-chambered heart table
9	Why do we call our heart myogenic?	15.5 - auto regulated by the nodal tissue itself
10	Sino-atrial node is called the pacemaker of our heart. Why?	15.3.1b - SAN generates the maximum action potentials and sets the rhythm
11	What is the significance of atrio-ventricular node and atrio-ventricular bundle in the functioning of heart?	15.3.1b + 15.3.2 - they conduct the action potential to the ventricular musculature
12	Define a cardiac cycle and the cardiac output.	15.3.2 and 15.3.2a
13	Explain heart sounds.	15.3.2b - lub and dub
14	Draw a standard ECG and explain the different segments in it.	15.3.3 + Figure 15.3 - P wave, QRS complex, T wave